

SHAFIN AHMED

Electrical & Electronic Engineer

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PROFESSIONAL SUMMARY

Electrical & Electronic Engineering graduate (CGPA 3.53/4.00, BRAC University) with hands-on experience in power systems, renewable energy, IoT, robotics, and semiconductor design. Proficient in AutoCAD-based electrical design, MATLAB/Simulink simulation, PCB layout, and computer vision pipelines. Completed industrial training at Ghorashal Combined Cycle Power Plant (BPDP) and Neural Semiconductor, gaining practical exposure to SCADA/DCS operations, switchgear, transformers, protection systems, and EDA workflows. Co-developed an IoT health-monitoring system that received a BDT 30,000 pre-seed grant. Adept at working in cross-functional teams on complex, real-world engineering projects.

SKILLS

Programming Languages: Python, C, MATLAB, Verilog, SystemVerilog

EDA & Design Tools: Altium Designer, Cadence Virtuoso, PSpice, Proteus, EasyEDA, LibrePCB

Simulation & Analysis: MATLAB, Simulink, AutoCAD (Electrical), Fusion 360 (3D / PCB)

Software & Platforms: VS Code, Google Workspace, Microsoft 365

Technical Domains: Power Systems, Renewable Energy, Smart Grid, IoT, Robotics, Computer Vision, PCB Design, Semiconductor / RFIC

Soft Skills: Analytical Thinking, Technical Writing, Team Collaboration, Problem Solving, Adaptability

EDUCATION

B.Sc. in Electrical & Electronic Engineering

BRAC University

CGPA: 3.53 / 4.00

2022 – 2026

Dhaka, Bangladesh

Higher Secondary Certificate (HSC)

Engineering University School & College

GPA: 5.00 / 5.00

2020

Dhaka, Bangladesh

Secondary School Certificate (SSC)

Sonargaon GR Institution

GPA: 5.00 / 5.00

2018

Narayanganj, Bangladesh

INTERNSHIP & INDUSTRIAL TRAINING

Industrial Training Program / *Ghorashal Power Plant Training Centre, BPDP*

Narsingdi, Bangladesh

Jan 2026

- Studied combined-cycle power plant operations covering turbines, boilers, substations, and industrial auxiliary systems.
- Gained practical exposure to DCS/SCADA systems, transformers, switchgear, and protection relay coordination.
- Participated in simulator-based training for real-time plant monitoring and fault response.

Industrial Attachment Program / *Neural Semiconductor*

Dhaka, Bangladesh

Sep – Oct 2025

- Gained hands-on experience in SystemVerilog, digital design, and analog IC design fundamentals.
- Explored RFIC concepts and industry-standard EDA workflows using Cadence tools.

PROJECTS

Final Year Design Project — 4-DOF Robotic Manipulator with Soft Gripper 2025 – 2026

- Designed and built a 4-DOF robotic arm for automated pick-and-place, incorporating a bio-inspired soft gripper.
- Developed a real-time computer vision system on Raspberry Pi 5 using OpenCV for object detection and localization.
- Implemented closed-loop control algorithms for precise, repeatable object handling.
- *Awarded 2nd Runner-Up at BRAC University Final Year Design Project Showcase (Jan 2026).*
- Project Link -  hdl.handle.net/10361/27527

BIOSE HealthTrack — IoT Smart Healthcare Monitoring System 2024 – 2025

- Received a **BDT 30,000 pre-seed grant** from the BRAC University Innovation Hub Program.
- Integrated multiple biomedical sensors with real-time data acquisition and CSV-based logging.
- Built a machine learning pipeline for early diabetic risk prediction from sensor data.
- Designed custom PCBs in Altium Designer and 3D enclosures in Fusion 360.

BRACU ALTER — Autonomous Underwater/Ground Vehicle 2023 – 2024

- Contributed to the AI & Control team; trained object-detection models with YOLOv5 and YOLOv8.
- Built a real-time operator GUI in PyQt5 with live OpenCV video streaming.
- Assisted in embedded control logic for autonomous navigation and obstacle avoidance.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Junior Executive, IT Department / Robotics Club of BRAC University (ROBU) 2022 – 2024

Managed digital platforms and supported technical infrastructure for club events.

Event Management Member / BRAC University Electrical & Electronic Club (BUEEC) 2022 – 2024

Co-organised workshops, seminars, and technical competitions for the EEE student community.

AWARDS & FUNDING

Pre-Seed Grant University Innovation Hub, BRAC University — BDT 30,000 for IoT Health Monitoring System Jul 2025

2nd Runner-Up Final Year Design Project Showcase, BRAC University Jan 2026

INTERESTS

Solar Energy Systems ▪ Renewable Energy ▪ Smart Grid ▪ Power Electronics ▪ Energy Storage Systems ▪ Power Systems ▪ Industrial Automation ▪ Embedded Systems ▪ Manufacturing Process Optimization ▪ Product Development ▪ Consumer Electronics Design ▪ Sustainable Technology

REFERENCE

Dr. Shameem Ahmad

Associate Professor, Dept. of Electrical & Electronic Engineering

 BRAC University

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